

LEEUWARDEN AB, Netherlands

WMO: 062700

Lat: 53.223N

Lon: 5.752E

Elev: 3

StdP: 14.69

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 19.8	(c) 23.1	(d) 13.4	(e) 10.9	(f) 22.0	(g) 17.9	(h) 13.5	(i) 24.4	(j) 33.1	(k) 45.8	(l) 30.1	(m) 45.3	(n) 8.6	(o) 80	(p) 0.616

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.8	81.0	68.3	76.8	66.5	73.3	64.7	69.8	78.4	67.7	74.8	65.8	71.6	8.8	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
66.8	98.5	73.7	64.9	92.3	71.0	63.4	87.4	69.1	33.8	78.4	32.1	74.8	30.6	71.8	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 27.0	(b) 23.5	(c) 20.8	(e) 13.8	(f) 88.5	(g) 6.6	(h) 2.7	(i) 9.1	(j) 90.4	(k) 5.3	(l) 92.0	(m) 1.6	(n) 93.5	(o) -3.2	(p) 95.4		
(a) 27.0	(b) 23.5	(c) 20.8	(e) 13.3	(f) 74.3	(g) 6.4	(h) 1.9	(i) 8.6	(j) 75.6	(k) 4.9	(l) 76.7	(m) 1.3	(n) 77.8	(o) -3.4	(p) 79.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	49.7	37.2	37.8	41.8	47.8	53.8	58.7	63.2	63.2	58.2	51.5	44.3	38.7
	DBStd	11.05	7.03	6.75	5.72	6.33	5.97	5.17	5.13	4.78	4.57	5.59	5.61	7.19
	HDD50	1725	398	343	259	115	27	0	0	0	1	48	182	352
	HDD65	5671	863	762	720	516	352	201	100	94	210	417	621	815
	CDD50	1631	0	0	4	49	145	262	408	408	247	96	11	1
	CDD65	101	0	0	0	0	4	12	42	37	6	0	0	0
	CDH74	685	0	0	0	6	31	111	284	220	33	0	0	0
	CDH80	150	0	0	0	0	2	24	70	50	4	0	0	0
Wind	WSAvg	10.1	11.9	11.5	11.7	10.0	9.9	9.3	8.7	8.3	8.9	9.8	10.1	11.1
Precipitation	PrecAvg	34.6	3.0	2.4	2.0	1.7	2.3	3.0	3.3	3.7	3.4	3.2	3.3	3.2
	PrecMax	43.2	5.7	6.2	3.9	4.1	5.0	5.6	6.9	8.7	7.6	7.7	6.1	6.9
	PrecMin	24.0	0.3	0.9	0.4	0.0	0.9	1.1	1.5	0.7	0.6	1.0	0.4	1.4
	PrecStd	5.2	1.4	1.1	1.1	0.9	1.1	1.3	1.5	2.0	1.7	1.4	1.3	1.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.1	53.8	61.5	73.4	78.2	83.4	86.4	85.5	78.6	68.9	59.2	54.2
		MCWB	50.1	48.2	52.2	58.7	62.5	69.2	70.5	69.1	66.6	62.0	56.4	52.3
	2%	DB	49.7	50.3	56.2	66.9	72.8	76.9	81.0	80.0	72.4	64.1	56.5	52.0
		MCWB	48.0	47.4	50.3	56.2	61.3	66.4	69.0	68.2	64.5	59.9	54.4	50.2
	5%	DB	47.8	48.4	52.4	62.2	68.6	71.7	76.5	75.2	68.4	61.2	54.2	50.0
		MCWB	46.2	46.2	48.3	54.2	59.5	63.5	67.2	66.3	62.2	58.1	52.1	48.5
	10%	DB	46.2	46.6	49.9	57.9	64.5	67.9	72.5	71.5	65.7	59.4	52.1	47.8
		MCWB	44.6	44.8	47.0	51.8	57.3	61.4	65.3	64.4	60.8	56.7	50.1	46.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.6	50.1	54.4	60.7	67.1	70.6	72.6	72.5	68.3	63.0	56.8	53.0
		MCDB	52.0	51.8	59.1	70.1	74.7	80.8	81.5	81.4	76.1	66.9	58.2	53.9
	2%	WB	48.3	48.0	51.1	57.3	62.9	67.1	70.1	69.2	65.2	60.4	54.8	50.8
		MCDB	49.3	49.7	54.7	65.1	70.0	74.8	78.7	76.6	70.8	63.4	56.3	51.9
	5%	WB	46.4	46.5	49.1	54.8	60.2	64.4	68.0	67.2	63.1	58.7	52.5	48.6
		MCDB	47.5	48.1	51.9	61.0	66.3	70.4	75.2	73.5	67.2	60.9	54.1	49.7
	10%	WB	44.7	44.9	47.2	52.7	57.9	62.1	65.9	65.4	61.5	57.1	50.4	46.6
		MCDB	45.9	46.4	49.6	57.2	63.8	66.9	71.4	70.5	65.0	59.2	51.8	47.7
Mean Daily Temperature Range	5% DB	MDBR	7.7	9.1	11.5	14.6	14.4	13.7	13.8	14.0	12.7	11.0	8.5	8.0
		MCDBR	7.7	10.2	15.8	20.9	21.3	20.8	20.6	19.6	16.5	12.3	8.8	7.5
		MCWBR	7.4	8.4	11.3	12.7	12.4	12.3	11.3	10.4	9.9	9.1	7.8	7.3
	5% WB	MCDBR	7.5	9.2	13.9	18.9	19.0	18.5	19.0	17.6	14.7	11.5	8.8	7.5
	MCWBR	7.5	8.0	10.6	12.1	12.0	11.7	11.1	10.3	9.5	9.0	8.1	7.5	
Clear-Sky Solar Irradiance	taub	0.330	0.348	0.376	0.404	0.403	0.401	0.412	0.405	0.388	0.362	0.342	0.322	
	taud	2.315	2.339	2.285	2.254	2.313	2.336	2.313	2.347	2.388	2.406	2.346	2.309	
	Ebn at Noon	201	237	254	262	269	270	264	259	249	229	194	179	
	Edh at Noon	19	26	33	38	38	38	38	35	30	24	19	16	
All-Sky Solar Radiation	RadAvg	192	398	815	1341	1664	1738	1664	1387	966	534	229	145	
	RadStd	18	66	79	128	111	115	173	105	86	49	21	16	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.67	N/A	N/A	N/A	N/A	N/A	N/A	-252	+97	N/A
(47) Regional (53 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page